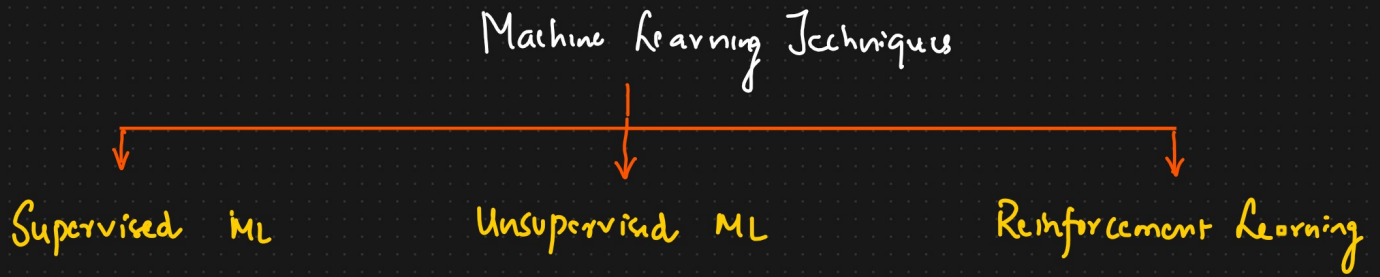


# 3 Types of Machine Learning



Supervised ML Technique

- Regression ✓
- Classification ✓

Predict House price

| Dataset       |                     |                          |
|---------------|---------------------|--------------------------|
| Size of House | # of Rooms          | Price                    |
|               | Independent feature | Dependent or o/p feature |
| 5000          | 5                   | 450K                     |
| 6000          | 6                   | 500K                     |
| —             | —                   | —                        |
| —             | —                   | —                        |

Continuous

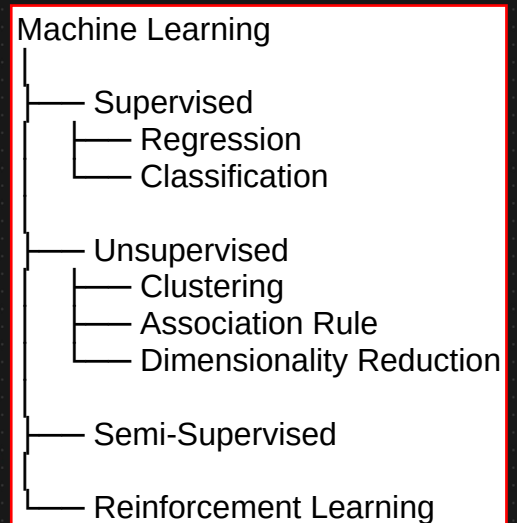
- ① Dependent or o/p feature
- ② Continuous → Regression
- ③ Categorical → Classification

Classification

| No. of study hours | No. of play hours | Pass/Fail |
|--------------------|-------------------|-----------|
| 7                  | 3                 | Pass      |
| 2                  | 6                 | Fail      |

Binary Classification OR Multiclass classification

May Be



② Unsupervised ML : Eg: Customer Segmentation → Clusters

Salary

Spending Score (1-10)

20000

9

45000

2

-

-

-

-

-

-

Spending



Ecommerce Company



Email with some discount to

Supervised ML

Unsupervised ML

① Linear Regression

② Ridge & Lasso

③ ElasticNet

④ Logistic Regression (Classification)

⑤ Decision Tree

⑥ Random Forest

⑦ AdaBoost

⑧ Xgboost

Both

Classification

& Regression

① K Means

② Hierarchical Mean

③ DBScan clustering

Reinforcement Learning